

① REQUIREMENTS — canonical enterprise-Q&A RAG workload

~2,000 users · ~10/40 rps · 9.2K-in/300-out · quality ~70B



② CONSTRAINTS — hard gates

memory (residency $\leq$ host) · latency (TTFT/TPOT) · economics (per-host ceiling)



③ GENERATE CANDIDATES

(a) 8×H100 + prefix cache · (b) P/D-disaggregated 2-pool · (c) KV-quantized 7B



④ TEST AGAINST memory · latency · economics

(a) 8×H100 + prefix cache

MEM ✓ · latency ✓ · cost ✓ | but prefill ~19.8K $\ll$ ~92K needed

✗ capacity (needs ~5 hosts)

(b) P/D-disaggregated 2-pool

MEM ✓ · latency ✓ · 2 hosts + fabric (~2× capex)

✓ survives as the at-scale answer

(c) KV-quantized 7B

MEM ✓ · cheapest (1 H100) | but fails the QUALITY gate

✗ EXCLUDED



⑤ OUTCOME — SURVIVORS

(a) per-request-latency baseline
(b) at-scale P/D answer — carried to Ch.13-14

✗ REJECTED

(c) fails quality gate
→ excluded from evaluation